Mediators of Inflammation

Lecture-based notes with extra detail. Source: Acute Inflammation lecture (Robbins 11th Edn), section (e), slides 64-86

How to read this: normal text and tables follow your lecturer's slides, in the same order (i) vasoactive amines, (ii) arachidonic acid metabolites, (iii) cytokines and chemokines, (iv) complement system, (v) other mediators. The **teal boxes** are extra detail that goes beyond the slides (mechanisms, clinical links, drugs), useful for understanding and for longer exam answers.

(e) MEDIATORS OF INFLAMMATION: OVERVIEW

Slides 64-66

Inflammatory mediators are substances that **start, amplify and regulate** inflammation.

Key mediators of acute inflammation

Group	Mediators
Vasoactive amines	Histamine, serotonin
Lipid mediators	Prostaglandins, leukotrienes (and lipoxins)
Cytokines and chemokines	TNF, IL-1, IL-6, IL-8
Complement fragments	C3a, C5a ("anaphylatoxins")

Where do mediators come from?

A. Cell-derived mediators	B. Plasma-derived mediators
Stored in granules → released rapidly. Example: histamine (mast cells). **Synthesised on demand** → slower release. Examples: prostaglandins, leukotrienes, cytokines. **Major cell sources:** macrophages, dendritic cells, mast cells (also platelets, neutrophils, endothelial cells). Different mediators have **short half-lives**, so inflammation is limited unless the stimulus persists.	**Made by the liver.** Circulate as **inactive precursors**. Activated by **proteolytic cascades** (e.g., the complement system). **Examples:** complement proteins (C3a, C5a, C3b), coagulation factors, kinin system (bradykinin).

Three general properties of mediators

Property	Explanation	Example
Produced only when needed	Released only after a stimulus (microbial products, substances from necrotic cells). Ensures inflammation occurs at the right time and place, and not when there is no threat.	Microbial products, necrotic cell contents
Very short-lived	Rapidly broken down, inactivated or removed. Prevents inflammation running out of control.	Bradykinin (kininase), C3a/C5a
One mediator triggers others	Activation of further mediators gives **amplification** of the response.	Complement fragments → histamine release; TNF → IL-1 and chemokine production

EXTRA DETAIL (beyond the slides)

- **Timeline:** pre-formed mediators (histamine, serotonin) act within **seconds to minutes**; plasma cascades (complement, kinins) within minutes; newly synthesised lipid mediators (prostaglandins, leukotrienes) within minutes to hours; **cytokines** need gene transcription, so they act over **hours** and sustain the response.
- **Mast cells and tissue macrophages are the "sentinels":** they sit in tissue next to vessels, detect microbes and damage through receptors (e.g., TLRs, toll-like receptors), and release the first wave of mediators.
- **Redundancy:** many mediators share the same effect (e.g., vasodilation by histamine, PGE_2, NO and bradykinin). Blocking one mediator rarely abolishes inflammation, which is why single-target drugs have limited effect in conditions like sepsis.
- **Two ways to stay in control:** mediators are short-lived, and **host cells carry regulators** (e.g., antiproteases, antioxidants, complement regulators) that limit tissue damage.

(i) VASOACTIVE AMINES: HISTAMINE AND SEROTONIN

Slides 67-68

Histamine

- One of the **earliest mediators** released in acute inflammation, because it is **pre-formed in granules**.
- **Major source: mast cells** (also basophils and platelets).
- **Main effects: vasodilation of arterioles** → redness and heat; **increased venular permeability** → oedema (exudate formation).
- **Key role: principal mediator of the immediate transient phase** of vascular leakage, by creating **interendothelial gaps** in venules.

Triggers for histamine release

- **Physical injury:** trauma, heat, cold
- **Allergic reactions:** antigen binding to **IgE on mast cells**
- **Complement fragments:** C3a, C5a (anaphylatoxins)
- **Neuropeptides:** e.g., substance P
- **Cytokines:** IL-1, IL-8

How histamine acts

- Binds **H1 receptors** on microvascular endothelial cells, causing **endothelial contraction**, **plasma leakage → oedema** and **vasodilation**.
- **Clinical relevance: antihistamines** block H1 receptors and prevent vascular effects in urticaria and allergies.
- Histamine also **contracts some smooth muscle** (e.g., bronchial smooth muscle → wheezing in asthma).

EXTRA DETAIL (beyond the slides)

- **Formation:** made from the amino acid **histidine** by **histidine decarboxylase**, then stored in mast cell and basophil granules bound to heparin and proteases.
- **Mast cell degranulation:** allergen cross-links IgE bound to **FcεRI** receptors, raising intracellular calcium and causing granules to fuse with the membrane and empty. C3a/C5a and substance P act through their own receptors on the mast cell.
- **Time course:** histamine leakage starts within minutes and is **brief (about 15-30 minutes)**, because histamine is quickly inactivated by **histaminase**. A later **delayed prolonged** leakage (hours) is driven by cytokines (TNF, IL-1) and direct injury.
- **Why venules?** Histamine acts mainly on venules because their endothelium carries the H1 receptors. Capillaries and arterioles are less affected, so leakage is concentrated where fluid and cells exit.
- **Other receptors: H2** receptors drive gastric acid secretion from parietal cells (blocked by ranitidine/famotidine-type drugs); this is separate from the vascular effect.
- **Anaphylaxis link:** massive systemic histamine release causes widespread vasodilation, leakage and bronchospasm (hypotension, laryngeal oedema); treated with adrenaline.

Serotonin (5-hydroxytryptamine, 5-HT)

- A **preformed vasoactive mediator**, mainly in **platelets**.
- Released when **platelets aggregate** (e.g., during clot formation).
- Causes **vasoconstriction** but has a **minor role compared to histamine**.
- Also produced by **neuroendocrine cells in the GIT** (gastrointestinal tract).
- Functions **predominantly as a neurotransmitter** in the GI tract and CNS (central nervous system).
- Only **limited involvement in acute inflammation**.

EXTRA DETAIL (beyond the slides)

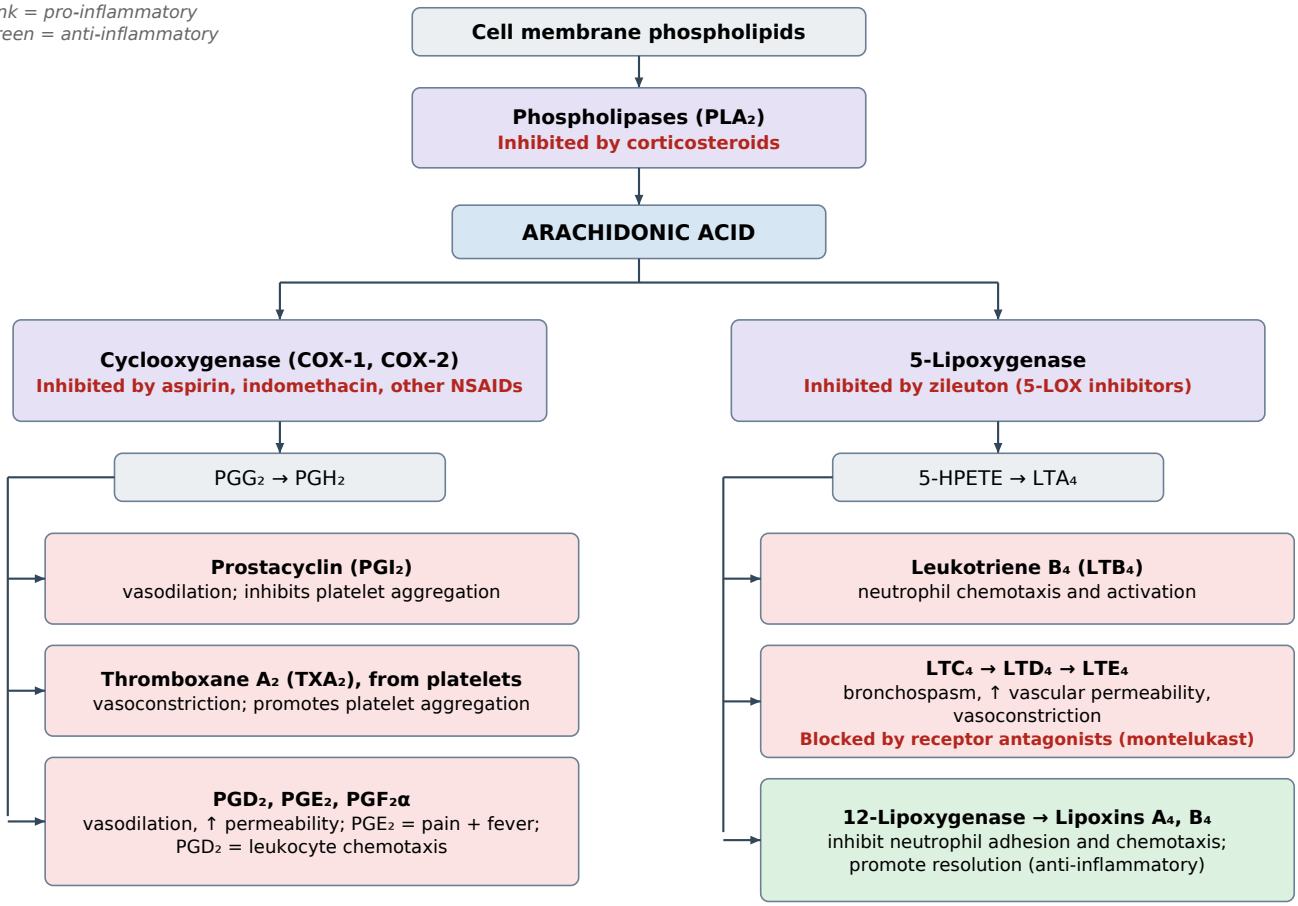
- **Formation:** synthesised from the amino acid **tryptophan** (via 5-hydroxytryptophan). Platelets do not make it: they **take it up** from plasma and store it in dense granules.
- **Most of the body's serotonin** (about 90%) is in **enterochromaffin cells** of the gut, which are the neuroendocrine cells the slide refers to.
- **Platelet release triggers:** collagen, thrombin, ADP, antigen-antibody complexes and PAF. This links **clotting to inflammation**.
- **Exam tip:** if asked for the main vasoactive amine in acute inflammation, answer **histamine**. Serotonin is the platelet-derived, vasoconstricting, minor one.

(ii) ARACHIDONIC ACID (AA) METABOLITES: PROSTAGLANDINS AND LEUKOTRIENES

Slides 69-72

AA metabolism gives **three major product groups**, made through **two major enzyme pathways** (cyclooxygenase and lipoxygenase).

Pink = pro-inflammatory
Green = anti-inflammatory



Production of arachidonic acid metabolites and their roles in inflammation (redrawn from the lecture figure). Red text marks the enzymes or receptors blocked by drugs. HPETE = hydroperoxyeicosatetraenoic acid.

EXTRA DETAIL: how AA metabolites are made

- **Formation of free AA:** AA is a **20-carbon polyunsaturated fatty acid** kept esterified in **membrane phospholipids**. Stimuli (injury, C5a, microbial products, cytokines) raise calcium and activate **phospholipase A₂ (PLA₂)**, which cuts AA free. This is the **rate-limiting first step**, so blocking it (steroids) shuts down *both* pathways.
- **Short-lived and local:** eicosanoids are not stored; they are made on demand, act on nearby cells through **G-protein-coupled receptors**, and are quickly broken down.
- **Other lipoxygenases** make HPETEs and HETEs (hydroxyeicosatetraenoic acids), shown as a side branch on the lecture figure.

1. Cyclooxygenase (COX) pathway → prostaglandins and thromboxane

- **COX-1:** constitutive ("housekeeping"). **COX-2:** induced during inflammation.

Product	Source / notes	Effects
PGE_2	Prostaglandin	**Fever, pain, vasodilation**
PGD_2 and $PGF_2\alpha$	Prostaglandins	**Vasodilation, ↑ permeability** (PGD_2 also leukocyte chemotaxis)
PGI_2 (prostacyclin)	Endothelium	**Vasodilation; inhibits platelet aggregation**
Thromboxane A_2 (TXA_2)	From platelets	**Vasoconstriction + platelet aggregation**

EXTRA DETAIL (beyond the slides)

- **Why COX-1 matters:** COX-1 makes the prostaglandins that **protect the stomach** (mucus, bicarbonate, blood flow) and support platelets and kidney perfusion. Blocking it explains **NSAID-induced gastric ulcers** and bleeding tendency.
- **Fever:** pyrogens (IL-1, TNF, IL-6) raise **PGE_2 in the hypothalamus**, which resets the temperature set point. Aspirin and NSAIDs lower fever by blocking COX-2-derived PGE_2.
- **Pain:** PGE_2 **sensitises nerve endings** to bradykinin and other stimuli, so tissue hurts more at a given stimulus (hyperalgesia).
- **PGI_2 versus TXA_2 balance:** endothelial PGI_2 keeps blood flowing (vasodilation, anti-platelet); platelet TXA_2 promotes clotting. Selective COX-2 inhibitors (e.g., celecoxib) cut PGI_2 without cutting platelet TXA_2, which is one reason they carry a **cardiovascular thrombotic risk**.
- **Low-dose aspirin** is anti-platelet because aspirin acetylates COX-1 **irreversibly**, and platelets (no nucleus) cannot make new COX for their 7-10 day life.

2. Lipoxygenase (LOX) pathway → leukotrienes and lipoxins

(a) Leukotrienes (pro-inflammatory)

- Produced by **neutrophils, macrophages and mast cells**.
- **LTB_4:** strong **neutrophil chemoattractant**.
- **LTC_4, LTD_4, LTE_4: bronchoconstriction**, **↑ vascular permeability**, **vasoconstriction**.

(b) Lipoxins (anti-inflammatory / pro-resolving)

- Generated by **neutrophils and platelets** (12/15-LOX pathway).
- **Lipoxin A_4 and B_4: inhibit neutrophil recruitment** and **promote resolution** of inflammation.

EXTRA DETAIL: leukotrienes and lipoxins

- **Transcellular biosynthesis of lipoxins:** neutrophils make the intermediate **LTA_4** and pass it to **platelets**, whose 12-lipoxygenase converts it into **LXA_4 and LXB_4**. Two cell types cooperate, and the product *switches inflammation off*.
- **The "class switch":** early inflammation is dominated by prostaglandins and leukotrienes; as it resolves, cells shift to making lipoxins and (from omega-3 fats) **resolvins**. This is a key signal for termination of the response.
- **Cysteinyl leukotrienes** (LTC_4, LTD_4, LTE_4) are hundreds of times more potent bronchoconstrictors than histamine and are central to **asthma**.
- **Aspirin-sensitive asthma:** in some asthmatics, aspirin blocks COX and shunts AA down the LOX pathway, producing more leukotrienes and triggering bronchospasm. **Aspirin** also acetylates COX-2 so that it produces pro-resolving "aspirin-triggered lipoxins".

Pharmacologic inhibitors of prostaglandins and leukotrienes

Eicosanoids play key roles in **vascular changes, pain, fever and leukocyte recruitment**, so many anti-inflammatory drugs block their synthesis or actions.

Drug class	Examples	Mechanism	Effects / use
1. COX inhibitors (NSAIDs)	Aspirin, ibuprofen, naproxen	Block COX-1 and COX-2 → ↓ prostaglandins. **Aspirin inhibits COX irreversibly.**	↓ Pain, ↓ fever, ↓ inflammation. **NSAIDs do not reduce leukotrienes** (LOX pathway unaffected).
2. 5-LOX inhibitors	Zileuton	Block leukotriene synthesis	Asthma (leukotrienes cause bronchoconstriction + oedema)
3. Leukotriene receptor antagonists	Montelukast, zafirlukast	Block leukotriene receptors (especially LTC_4/LTD_4)	Asthma; allergic rhinitis
4. Corticosteroids	Prednisolone, dexamethasone	↓ PLA_2 → ↓ AA release; ↓ COX-2 expression; ↓ pro-inflammatory cytokines (IL-1, TNF); ↓ iNOS and many inflammation-related genes	Very powerful suppression of inflammation. Use: autoimmune disease, severe asthma, allergic reactions.

Drug class	Examples	Mechanism	Effects / use
5. Dietary lipids (fish oil, omega-3)	Omega-3 fatty acids	Poor substrates for COX and LOX → less inflammatory eicosanoids; promote anti-inflammatory mediators (e.g., resolvins)	Adjunct in chronic inflammatory states

EXTRA DETAIL (beyond the slides)

- **Steroid mechanism in more detail:** glucocorticoids bind an intracellular receptor, move to the nucleus, and **repress transcription factors such as NF-κB** (nuclear factor kappa B) while inducing anti-inflammatory proteins (e.g., annexin A1/lipocortin, which inhibits PLA_2). That is why they affect so many mediators at once.
- **Why steroids work better than NSAIDs for severe inflammation:** NSAIDs only block the COX arm, but steroids block the step *above* both arms and also cytokine genes.
- **Side effects to know:** NSAIDs → gastric ulcer, renal impairment, bleeding; steroids → immunosuppression and infection risk, hyperglycaemia, osteoporosis, Cushingoid features.

(iii) CYTOKINES AND CHEMOKINES

Slides 73-79

Cytokines are small protein messengers produced by many cells that **mediate and regulate immune and inflammatory responses**.

Who produces cytokines?	What do cytokines do?
Main producers: activated lymphocytes, macrophages, dendritic cells. **Other contributors:** endothelial cells, epithelial cells, fibroblasts / connective tissue cells.	• Coordinate inflammation • Recruit and activate leukocytes • Regulate the immune response • Influence healing and tissue repair

TNF (tumour necrosis factor) and IL-1 (interleukin-1): the key cytokines

Who produces them?	What triggers their release?
Mainly **activated macrophages and dendritic cells**. TNF also from T cells and mast cells. IL-1 also from epithelial cells.	Microbial products (via TLRs), dead cells, immune complexes, foreign bodies, physical injury. **IL-1 activation requires the inflammasome.**

Major actions of TNF and IL-1

- **1. Endothelial activation (most important function):** ↑ adhesion molecules (**E-selectin, P-selectin, ICAM-1, VCAM-1**), ↑ chemokines and cytokines, ↑ eicosanoids (PGs, LTs), ↑ procoagulant activity. **Outcome:** promotes leukocyte adhesion → migration → inflammation.
- **2. Activation of leukocytes and local cells:** TNF increases neutrophil responses and activates macrophages (↑ NO production); IL-1 stimulates fibroblast collagen synthesis; IL-1 promotes Th17 responses, strengthening acute inflammation.
- **3. Systemic acute-phase responses:** TNF, IL-1 and IL-6 mediate **fever**, **acute-phase protein production** and **leukocytosis**. TNF also affects metabolism: ↑ lipid and protein breakdown, ↓ appetite; **chronic TNF → cachexia** (chronic infections, cancers).

Clinical correlations

- **TNF antagonists:** very effective in **rheumatoid arthritis, psoriasis and inflammatory bowel disease**, but ↑ risk of **tuberculosis and intracellular infections** (macrophages rely on TNF to kill intracellular microbes).
- **IL-1 antagonists:** less effective than TNF blockers (exact reason unclear).
- **Sepsis:** blocking TNF/IL-1 does **not** improve survival (many other cytokines are involved).

Cytokines in inflammation (lecture table)

Cytokine	Principal sources	Principal actions in inflammation
In acute inflammation		

Cytokine	Principal sources	Principal actions in inflammation
TNF	Macrophages, mast cells, T lymphocytes	Stimulates expression of endothelial adhesion molecules and secretion of other cytokines; systemic effects
IL-1	Macrophages, endothelial cells, some epithelial cells	Similar to TNF; greater role in fever
IL-6	Macrophages, other cells	Systemic effects (acute-phase response)
Chemokines	Macrophages, endothelial cells, T lymphocytes, mast cells, other cell types	Recruitment of leukocytes to sites of inflammation; migration of cells in normal tissues
IL-17	T lymphocytes	Recruitment of neutrophils and monocytes
In chronic inflammation		
IL-12	Dendritic cells, macrophages	Increased production of IFN-γ
IFN-γ	T lymphocytes, NK cells	Activation of macrophages (increased ability to kill microbes and tumour cells)
IL-17	T lymphocytes	Recruitment of neutrophils and monocytes

The slide notes that many other cytokines play lesser roles, and that cytokines of acute inflammation may also contribute to chronic inflammation.

Major roles of cytokines in acute inflammation (lecture figure)

Local inflammation	Systemic protective effects	Systemic pathological effects
TNF, IL-1 act on endothelial cells → ↑ expression of adhesion molecules; release of IL-1 and chemokines. **TNF** → ↑ vascular permeability. **TNF, IL-1** activate leukocytes → they release IL-1, IL-6 and chemokines.	**Brain** (TNF, IL-1, IL-6): **fever**. **Liver** (IL-1, IL-6): **acute-phase proteins**. **Bone marrow** (TNF, IL-1, IL-6): **leukocyte production**.	**TNF** on heart: **low cardiac output**. **TNF** on endothelium/vessels: **thrombosis** and ↑ **permeability**. **TNF** on skeletal muscle and other tissues: **insulin resistance**.

EXTRA DETAIL: cytokines

- **IL-1 and the inflammasome:** IL-1β is made as an **inactive precursor (pro-IL-1β)**. The **inflammasome** (e.g., **NLRP3**) senses microbes, crystals (urate, cholesterol), ATP and K^+ efflux, then activates **caspase-1**, which cuts pro-IL-1β into active IL-1β. This is the link to **gout** and the reason **IL-1 blockers** (anakinra, canakinumab) are used in gout and autoinflammatory syndromes.
- **Fever:** IL-1, TNF and IL-6 act on the hypothalamus to raise **PGE_2**, resetting the thermostat.
- **Acute-phase proteins (made by the liver under IL-6): CRP** (C-reactive protein, an opsonin that binds microbes and activates complement), **fibrinogen**, **serum amyloid A** and **hepcidin**. Fibrinogen makes red cells stack (rouleaux) → **raised ESR** (erythrocyte sedimentation rate). Hepcidin traps iron in macrophages → **anaemia of chronic disease**.
- **Leukocytosis:** IL-1, IL-6 and TNF push the bone marrow to release neutrophils early (left shift) and increase production through colony-stimulating factors.
- **Septic shock:** very high TNF (with IL-1) causes widespread vasodilation, increased permeability, **DIC** (disseminated intravascular coagulation), hypoglycaemia and cardiovascular failure.
- **Anti-TNF drugs:** infliximab, adalimumab (antibodies) and etanercept (soluble receptor). Screen for latent TB before starting, because TNF holds granulomas together.
- **IL-12 / IFN-γ axis:** IL-12 drives Th1 cells to make IFN-γ, which activates macrophages (M1). Defects in this axis cause severe mycobacterial infections.

Chemokines

- **Small cytokines that direct leukocyte movement** (chemoattractants): they tell leukocytes where to go.
- **Classified by the pattern of cysteine (C) residues** into four major classes.

Class	Examples	Main targets / notes
1. CXC (α-chemokines)	**IL-8** (produced by macrophages and endothelial cells)	Acts mainly on **neutrophils** → activation + chemotaxis

Class	Examples	Main targets / notes
2. CC (β-chemokines)	MCP-1, eotaxin, MIP-α, RANTES	Attract **monocytes, eosinophils, basophils, lymphocytes**. Do **NOT** attract neutrophils
3. C (γ-chemokines)	Lymphotactin	Attracts **lymphocytes**
4. CX3C	Fractalkine	Membrane-bound and soluble forms; attracts monocytes and T cells; mediates adhesion

Functions of chemokines

- **1. Acute inflammation:** produced in response to microbes, necrotic cells and cytokines. They **increase integrin affinity** on leukocytes (stronger adhesion) and **guide leukocytes by chemotaxis** to the infection/injury site.
- **2. Maintenance of tissue architecture:** "homeostatic chemokines" **arrange T cells and B cells in lymph nodes and spleen** and maintain normal immune structure.

EXTRA DETAIL (beyond the slides)

- **Alternative names (useful in papers):** IL-8 = **CXCL8**; MCP-1 = **CCL2** (monocyte chemoattractant protein 1); eotaxin = **CCL11** (eosinophils, in allergy); RANTES = **CCL5**; MIP-1α = **CCL3**; fractalkine = **CX3CL1**.
- **Receptors:** all chemokine receptors are **G-protein-coupled receptors**. Binding triggers actin polymerisation (leukocyte movement) and conformational activation of integrins (LFA-1, Mac-1).
- **Immobilised on endothelium:** chemokines bind **proteoglycans on the endothelial surface**, so rolling leukocytes meet a high local concentration. This is the "activation" step between rolling and firm adhesion.
- **Homeostatic examples:** CXCL13 organises B cells into follicles; CCL19/CCL21 draw T cells and dendritic cells into lymph node T-cell zones.
- **HIV link: CCR5** and **CXCR4** are chemokine receptors used by HIV as co-receptors for entry into T cells.

(iv) COMPLEMENT SYSTEM

Slides 80-83

Overview

- Complement = **plasma proteins that help defend against microbes** and contribute to inflammation.
- Produced **mainly in the liver**; circulate as **inactive precursors**.
- Activated → **enzymatic cascade** → strong amplification.

Activation pathways (all generate C3 convertase → cleaves C3)

Pathway	Trigger	Notes
1. Classical	**Antigen-antibody complexes** (IgM, IgG)	C1 binds antibody → cascade starts
2. Alternative	**Microbial surface molecules** (LPS, endotoxin)	Does **NOT** require antibody
3. Lectin	**Mannose-binding lectin (MBL)** binding to microbial carbohydrates	Activates C1-like proteins

EXTRA DETAIL: the three pathways

- **Classical:** C1 = C1q + C1r + C1s. C1q binds the Fc region of antibody (and CRP); activated C1s cleaves **C4 and C2** → **C4b2a**, the C3 convertase.
  Lectin: MBL (or ficolins) bound to **MASPs** (MBL-associated serine proteases); **MASP-2** cleaves C4 and C2 → the **same C4b2a**. The classical and lectin pathways differ only in the trigger.
- **Alternative:** C3 is constantly hydrolysed at a low rate ("**tick-over**") to C3(H_2O). With factor B and factor D this forms a fluid-phase convertase, which makes the first C3b. On a microbe, C3b + Bb = **C3bBb** (stabilised by **properdin**), which makes more C3b: an **amplification loop**.
- **Naming note:** some textbooks call the large C2 fragment **C2a** (so the convertase is C4b2a), others call it C2b. The function is the same: the large fragment joins C4b and cleaves C3.
- **Why microbes, not host cells:** host cells carry regulators (DAF, MCP, factor H binding via sialic acid) that destroy C3b; microbes lack them, so the loop runs away on microbes only.

Key step: C3 cleavage and main functions

- **C3 convertase splits C3 → C3a + C3b.** C3a = inflammatory mediator; C3b = binds the microbe → **opsonin**.
- **C3b helps form C5 convertase** → cleaves C5 → **C5a** (inflammatory) + **C5b** (starts membrane attack complex formation).

Main function	Mediators	Details
1. Inflammation	**C3a, C4a, C5a** (anaphylatoxins)	Induce **mast cell degranulation**, ↑ vascular permeability. **C5a** = strong chemotactic factor for neutrophils, monocytes and eosinophils; C5a also **activates AA metabolism** → more inflammatory mediators.
2. Opsonisation	**C3b and iC3b**	Coat microbes → phagocytes (with C3b receptors) bind and engulf more easily
3. Cell lysis	**C5b-C9 = MAC** (membrane attack complex)	Creates pores → osmotic lysis. **Especially important for *Neisseria*** (deficiency → recurrent *Neisseria* infections)

EXTRA DETAIL (beyond the slides)

- **Anaphylatoxin potency:** C5a > C3a > C4a. They are quickly inactivated by **carboxypeptidase N**, which removes the C-terminal arginine.
- **MAC assembly:** C5b binds C6, C7 (inserts into membrane), C8, then 10-16 molecules of C9 polymerise into a ring-shaped pore. It kills Gram-negative organisms efficiently but not Gram-positives, whose thick wall blocks access to the membrane.
- **Other roles of C3b:** clears immune complexes (via CR1 on red cells, carried to spleen and liver) and, as C3d, lowers the B-cell activation threshold through CR2.
- **Lab tests:** low C3 and C4 suggest classical-pathway consumption (e.g., lupus); low C3 with normal C4 suggests alternative-pathway activation. **CH50** measures total classical activity.

Complement regulation (prevents host damage)

Normal cells express regulatory proteins to stop excessive activation.

Regulator	Action	Deficiency / mutation
C1 inhibitor (C1-INH)	Blocks classical pathway activation	**Hereditary angioedema**
DAF and CD59 (GPI-anchored)	**DAF**: prevents C3 convertase formation. **CD59**: prevents MAC assembly	GPI-anchor deficiency → excessive complement activation → **paroxysmal nocturnal haemoglobinuria (PNH)**
Factor H	Inhibits alternative pathway; promotes breakdown of C3b	Mutation → **atypical haemolytic uraemic syndrome (HUS)**

Clinical relevance

- **Excess complement activation:** occurs in **autoimmune diseases** (antibodies deposit on host tissue → complement → inflammation and damage).
- **Complement deficiencies:** early components (**C1, C4**) → ↑ **immune complex disease**; **C3** deficiency → severe recurrent infections; **MAC (C5-C9)** deficiency → ***Neisseria*** infections.

EXTRA DETAIL: regulation and disease

- **C1-INH in more detail:** it also blocks the **lectin pathway (MASPs)**, **kallikrein** and **factor XIIa**. Without it, uncontrolled kallikrein makes excess **bradykinin**, causing attacks of swelling (angioedema) without itching or urticaria. Complement C4 is low even between attacks.
- **PNH:** an acquired *PIGA* mutation in a stem cell removes GPI anchors, so red cells lose DAF and CD59 and are lysed by complement → haemolysis, dark urine in the morning, thrombosis. Treated with **eculizumab** (anti-C5).
- **MBL deficiency:** common and usually silent, but may raise infection risk in infants and immunocompromised patients.
- **Therapy note:** anti-C5 therapy blocks MAC formation, so patients need **meningococcal vaccination** (link to *Neisseria*).

(v) OTHER MEDIATORS OF INFLAMMATION

Slides 84-86

Platelet-activating factor (PAF)

- Produced by **platelets, basophils, mast cells, neutrophils, macrophages and endothelial cells**.
- Originally named for causing **platelet aggregation**.
- A powerful inflammatory mediator with multiple effects: **vasodilation**; **↑ vascular permeability** (very potent, may exceed histamine); **bronchoconstriction**; **leukocyte activation**; **platelet activation**.

EXTRA DETAIL (beyond the slides)

- **Formation:** a **phospholipid-derived** mediator. PLA_2 cuts membrane phospholipid to give lyso-PAF, which is then acetylated to PAF.
- **Potency:** as a vasodilator and permeability factor it is reported to be roughly **100-10,000 times more potent than histamine** (low concentration), and at high concentration it constricts vessels and bronchi.
- **Receptor:** acts through a G-protein-coupled **PAF receptor**; it also stimulates synthesis of other mediators (e.g., eicosanoids), amplifying inflammation.

Products of coagulation

- **Coagulation and inflammation are closely linked.**
- **Protease-activated receptors (PARs)** on many cells are activated by clotting factors (e.g., **thrombin**). Major role: platelet activation.
- **Fibrin split products (fibrinopeptides):** promote inflammation; can **increase vascular permeability** and **attract leukocytes**.
- In tissue injury, clotting and inflammation often occur together → difficult to separate cause from effect.

EXTRA DETAIL (beyond the slides)

- **Thrombin** also acts on endothelium through PARs: ↑ P-selectin, chemokines, adhesion molecules, NO, PGI_2 and PAF, and promotes leukocyte adhesion and fibroblast proliferation.
- **Fibrinolysis:** plasmin lyses fibrin, but also **cleaves C3 → C3a** and **activates factor XII**, which feeds back into the kinin and clotting systems.
- **Hageman factor (factor XII) is the hub:** contact with exposed collagen or basement membrane activates factor XII, which switches on **the clotting, kinin and fibrinolytic systems** (and complement is linked through plasmin and C3). Tissue injury therefore turns on several plasma cascades together.

Kinins

- Kinins are **vasoactive peptides** produced from plasma proteins (**kininogens**), generated by **kallikreins** (a group of proteases).
- **Bradykinin:** formed when **kallikrein cleaves high-molecular-weight kininogen (HMWK)**.
- **Major effects (similar to histamine):** ↑ vascular permeability, vasodilation, smooth muscle contraction, and **pain** (important mediator of inflammatory pain).

- **Short-acting** → rapidly broken down by **kininase**.

EXTRA DETAIL (beyond the slides)

- **Pathway:** factor XIIa → prekallikrein → **kallikrein** → HMWK → **bradykinin**. Kallikrein also activates more factor XII (positive feedback).
- **Kininase = ACE** (angiotensin-converting enzyme) among others. **ACE inhibitors** let bradykinin build up → **dry cough and angioedema**.
- **Hereditary angioedema** (C1-INH deficiency) = overactive kallikrein and excess bradykinin, linking the complement and kinin systems.

Neuropeptides

What are they? / Where?	Examples	Why they matter in inflammation
Small peptide molecules released by **sensory nerves and leukocytes**. Help initiate and regulate inflammation. Produced in central and peripheral nervous systems; nerve fibres rich in neuropeptides are abundant in the **lungs and GI tract**.	**Substance P:** major neuropeptide in inflammation; transmits pain signals; ↑ vascular permeability; contributes to **neurogenic inflammation**. **Neurokinin A:** works similarly to substance P; supports inflammatory signalling pathways.	Leukocytes have receptors for neuropeptides → **cross-talk between nervous and immune systems**. Neural signals can modulate inflammation. **Example:** vagus nerve activation reduces release of pro-inflammatory cytokines (e.g., TNF) → the "**anti-inflammatory reflex**", explored in vagus nerve stimulation trials in rheumatoid arthritis.

EXTRA DETAIL (beyond the slides)

- **Neurogenic inflammation:** sensory nerve endings release substance P, which causes **mast cell degranulation**, vasodilation and leakage. This is why a scratch produces a "flare" (spreading redness) beyond the injury (the axon reflex).
- **Capsaicin** (chilli pepper) first releases and then depletes substance P, which is why topical capsaicin is used for neuropathic pain.
- **Anti-inflammatory reflex:** vagal nerve endings release **acetylcholine**, which acts on α7 nicotinic receptors on macrophages to reduce TNF release.

SUMMARY: WHICH MEDIATOR DOES WHAT?

Revision tables

Effect	Mediators
Vasodilation	Histamine, PGE_2, PGD_2, PGI_2, bradykinin, PAF (and NO)
Vasoconstriction	Serotonin, TXA_2, $LTC_4/LTD_4/LTE_4$
↑ Vascular permeability	Histamine, C3a, C5a, bradykinin, $LTC_4/LTD_4/LTE_4$, PAF, substance P, fibrinopeptides, PGD_2, TNF (delayed)
Endothelial activation / adhesion	TNF, IL-1 (↑ E-selectin, P-selectin, ICAM-1, VCAM-1); chemokines (↑ integrin affinity)
Chemotaxis / leukocyte recruitment	C5a, LTB_4, IL-8 (CXCL8) and other chemokines, PGD_2, fibrinopeptides
Fever	IL-1, TNF, IL-6 (via PGE_2)
Pain	PGE_2, bradykinin, substance P
Bronchoconstriction	Histamine, $LTC_4/LTD_4/LTE_4$, PAF
Acute-phase proteins / leukocytosis	IL-6, IL-1, TNF
Opsonisation	C3b, iC3b (and antibody, CRP)
Lysis of microbes	MAC (C5b-C9)
Resolution (anti-inflammatory)	Lipoxins (and resolvins from omega-3 fats)

Mediator group	Origin	Time to act
Histamine, serotonin	Pre-formed (mast cells, platelets)	Seconds to minutes
Complement, kinins, clotting products	Plasma (liver), activated by cleavage	Minutes
Prostaglandins, leukotrienes, PAF	Synthesised on demand from membrane phospholipid	Minutes to hours
Cytokines and chemokines	Synthesised on demand (new gene expression)	Hours

Memory lines
• **Histamine** = first, pre-formed, mast cells, H1, venular leakage. **Serotonin** = platelets, vasoconstriction, minor role.
• **COX → prostaglandins + TXA_2**; **5-LOX → leukotrienes**; **12-LOX → lipoxins**. Aspirin/NSAIDs block COX; zileuton blocks 5-LOX; montelukast blocks the leukotriene receptor; steroids block PLA_2.
• **TNF and IL-1** = endothelial activation + fever; **IL-6** = acute-phase proteins; **IL-8** = neutrophils; CC chemokines do **not** attract neutrophils.
• **C3b** = opsonin; **C3a/C4a/C5a** = anaphylatoxins; **C5a** = strongest chemoattractant; **MAC** = *Neisseria*.
• **Bradykinin** = pain + permeability; **PAF** = most potent permeability factor; **substance P** = pain + neurogenic inflammation.

Quick self-test

Question	Answer
Which mediator causes the immediate transient phase of vascular leakage?	Histamine (H1 receptors on venular endothelium)
Why does aspirin reduce fever and pain but not leukotriene-mediated asthma?	It blocks COX (prostaglandins) but not 5-LOX (leukotrienes)
Which cytokines mediate the acute-phase response?	TNF, IL-1 and IL-6
Why are anti-TNF drugs risky for TB?	Macrophages need TNF to kill intracellular microbes and maintain granulomas
Which chemokine class attracts neutrophils?	CXC (IL-8); CC chemokines do not
Which complement fragments are anaphylatoxins, and what do they do?	C3a, C4a, C5a: mast cell degranulation and ↑ vascular permeability
What deficiency causes recurrent Neisseria infections?	MAC components (C5-C9)
Which regulator deficiency causes hereditary angioedema?	C1 inhibitor (excess bradykinin)
Which regulators are missing in PNH?	DAF and CD59 (GPI-anchored)
Name two mediators of inflammatory pain.	Bradykinin and PGE_2 (also substance P)